

SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
COMPUTER SCIENCE & INFORMATION TECHNOLOGY DEPARTMENT

Spring 2025
Cloud Computing (CS-492T)
Assignment # 1

Semester: 6th
Announced Date:14-04-25
Total Marks: 3
Instructor Name: Engr. Tehmeed Ebaduddin

Batch: 2022F-BCS-152
Due Date:21-04-25
Marks Obtained:
Name:

CLO #	Course Learning Outcomes (CLOs)	PLO Mapping	Bloom’s Taxonomy
CLO 1	Describe the core concepts of the cloud computing paradigm	PLO_1 (Academic Education)	C1 (Knowledge)

Q1. **Write** down the steps to create a VM using one of the major Cloud platforms, such that the name of resource group and VM should reflect your name. Attach the configuration snapshot and attach the snapshot of the host machine connecting to the cloud VM.

Steps to Create a VM on Azure

1. Login to Azure Portal

- Go to <https://portal.azure.com>
- Sign in with your Azure account.

2. Create a Resource Group

- In the search bar, type **Resource Groups** and click it.
- Click on + **Create**.
 - **Subscription:** Select your subscription.
 - **Resource Group Name:** e.g., Faisal
 - **Region:** Choose a region close to your location.
- Click **Review + Create** → **Create**.

3. Create the Virtual Machine

- In the search bar, type **Virtual Machines** and click it.
- Click on + **Add** → + **Virtual machine**.
- Configure the following:

Basics:

- **Subscription:** Same as your resource group.
- **Resource Group:** Faisal
- **Virtual machine name:** Faisal
- **Region:** Same as the resource group.
- **Availability Options:** Leave default.
- **Image:** Select Ubuntu Server 22.04 LTS (or Windows, depending on your preference).
- **Size:** Choose a basic tier like B1s or D2s_v3 for testing.
- **Authentication type:** Password or SSH.
- **Username:** yourname
- **Password/SSH Key:** Set as needed.

Inbound Port Rules:

- **Public Inbound Ports:** Allow selected ports
- **Select inbound ports:** SSH (port 22) for Linux / RDP (port 3389) for Windows
- Click **Next** through other tabs or keep default settings.
- Click **Review + Create** → **Create**.

4. Wait for Deployment

- It will take a few minutes.
- Once deployed, go to the VM overview page.

5. Connect to VM

- Click **Connect**:
 - For **Linux (SSH)**:
 - Copy the public IP from Azure VM overview.
 - Open terminal on your local machine:
 - `ssh yourname@<public-ip>`
 - For **Windows (RDP)**:
 - Download the `.rdp` file.
 - Open it and enter your credentials.

✓ Configuration

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

CS22f-152@ssuet.edu.pk
SSUET (SSUET.EDU.PK)

Home > Virtual machines >

Create a virtual machine

...

Validation passed

Help me create a low cost VM

Help me create a VM optimized for high availability

Help me choose the right VM size for my workload

Basics

Subscription

Resource group

Virtual machine name

Region

Availability options

Zone options

Availability zone

Security type

Enable secure boot

Enable vTPM

Integrity monitoring

Image

Azure for Students

(new) Faisal

Faisal

East Asia

Availability zone

Self-selected zone

2

Trusted launch virtual machines

Yes

Yes

No

Windows Server 2022 Datacenter: Azure Edition Hotpatch - Gen2

< Previous

Next >

Create

Download a template for automation

Give feedback

SSUET

Sign out

CS22f-152@ssuet.e...

CS22f-152@ssuet.edu.pk

View account

Switch directory

...

Sign in with a different account

Basics

Subscription

Resource group

Virtual machine name

Region

Availability options

Zone options

Availability zone

Security type

Enable secure boot

Enable vTPM

Integrity monitoring

Image

VM architecture

Azure for Students

(new) Faisal

Faisal

East Asia

Availability zone

Self-selected zone

2

Trusted launch virtual machines

Yes

Yes

No

Windows Server 2022 Datacenter: Azure Edition Hotpatch - Gen2

x64

Enable Hibernation

Username

Public inbound ports

Already have a Windows license?

Azure Spot

No

2022F-BCS-152

RDP

No

No

Disks

OS disk size

OS disk type

Use managed disks

Delete OS disk with VM

Ephemeral OS disk

Image default

Premium SSD LRS

Yes

Enabled

No

Networking

Virtual network

Subnet

Public IP

Accelerated networking

Place this virtual machine behind an existing load balancing solution?

Delete public IP and NIC when VM is deleted

(new) Faisal-vnet

(new) default (10.0.0.0/24)

(new) Faisal-ip

On

No

Disabled

Management

Microsoft Defender for Cloud	Basic (free)
System assigned managed identity	Off
Login with Microsoft Entra ID	Off
Auto-shutdown	Off
Backup	Disabled
Site Recovery	Disabled
Enable periodic assessment	Off
Enable hotpatch	On
Patch orchestration options	Azure-orchestrated patching (preview): patches will be installed by Azure
Reboot setting	Reboot if required

Monitoring

Alerts	Off
Boot diagnostics	On
Enable OS guest diagnostics	Off
Enable application health monitoring	Off

Advanced

Extensions	None
VM applications	None
Cloud init	No
User data	No
Disk controller type	SCSI
Proximity placement group	None
Capacity reservation group	None

Microsoft Azure

Search resources, services, and docs (G+/f)

Copilot

CS22f-152@ssuet.edu.pk

Home >

CreateVm-MicrosoftWindowsServer.WindowsServer-202-20250421010000 | Overview

Deployment

Search

DeleteCancelRedeployDownloadRefresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name: CreateVm-MicrosoftWindowsServer.WindowsSe... Start time: 4/21/2025, 1:07:25 AM

Subscription: Azure for Students Correlation ID: 0444ac83-5720-413b-b172-078

Resource group: Faisal

Deployment details

Next steps

Setup auto-shutdown Recommended

Monitor VM health, performance and network dependencies Recommended

Run a script inside the virtual machine Recommended

Go to resourceCreate another VM

Give feedback

Tell us about your experience with deployment

Deployment succeeded

Deployment 'CreateVm-MicrosoftWindowsServer.WindowsServer-202-20250421010000' to resource group 'Faisal' was successful.

Go to resourcePin to dashboard

Cost Management

Get notified to stay within your budget and prevent unexpected charges on your bill.

Set up cost alerts >

Microsoft Defender for Cloud

Secure your apps and infrastructure

Go to Microsoft Defender for Cloud >

Free Microsoft tutorials

Start learning today >

Work with an expert

Azure experts are service provider partners who can help manage your assets on Azure

Microsoft Azure

Search resources, services, and docs (G+/f)

Copilot

CS22f-152@ssuet.edu.pk

Home > CreateVm-MicrosoftWindowsServer.WindowsServer-202-20250421010000 | Overview >

Faisal

Virtual machine

Search

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Connect

Networking

Settings

Availability + scale

Security

Backup + disaster recovery

Operations

Monitoring

Faisal virtual machine agent status is not ready. Troubleshoot the issue →

Help me copy this VM in any region

ConnectStartRestartStopHibernateCaptureDeleteRefreshOpen in

Essentials

Resource group (move) : Faisal

Status : Running

Location : East Asia (Zone 2)

Subscription (move) : Azure for Students

Subscription ID : 580cdf0a-6029-4043-8488-ad5e336fe2f0

Availability zone : 2

Tags (edit) : Add tags

Operating system : Windows

Size : Standard D2s v3 (2 vcpus, 8 GiB memory)

Public IP address : 20.255.56.118

Virtual network/subnet : Faisal-vnet/default

DNS name : Not configured

Health state : -

Time created : 4/20/2025, 8:07 PM UTC

PropertiesMonitoringCapabilities (8)RecommendationsTutorials

Virtual machine

Networking

SSUET

Sign out

CS22f-152@ssuet.e...

CS22f-152@ssuet.edu.pk

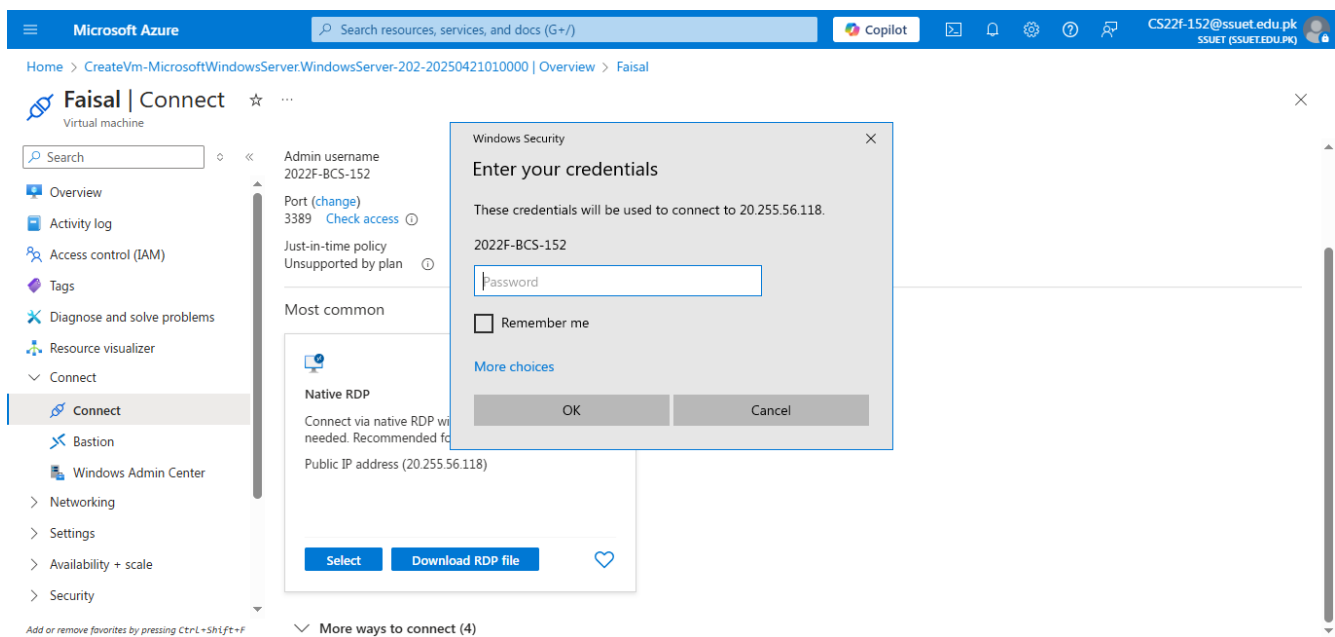
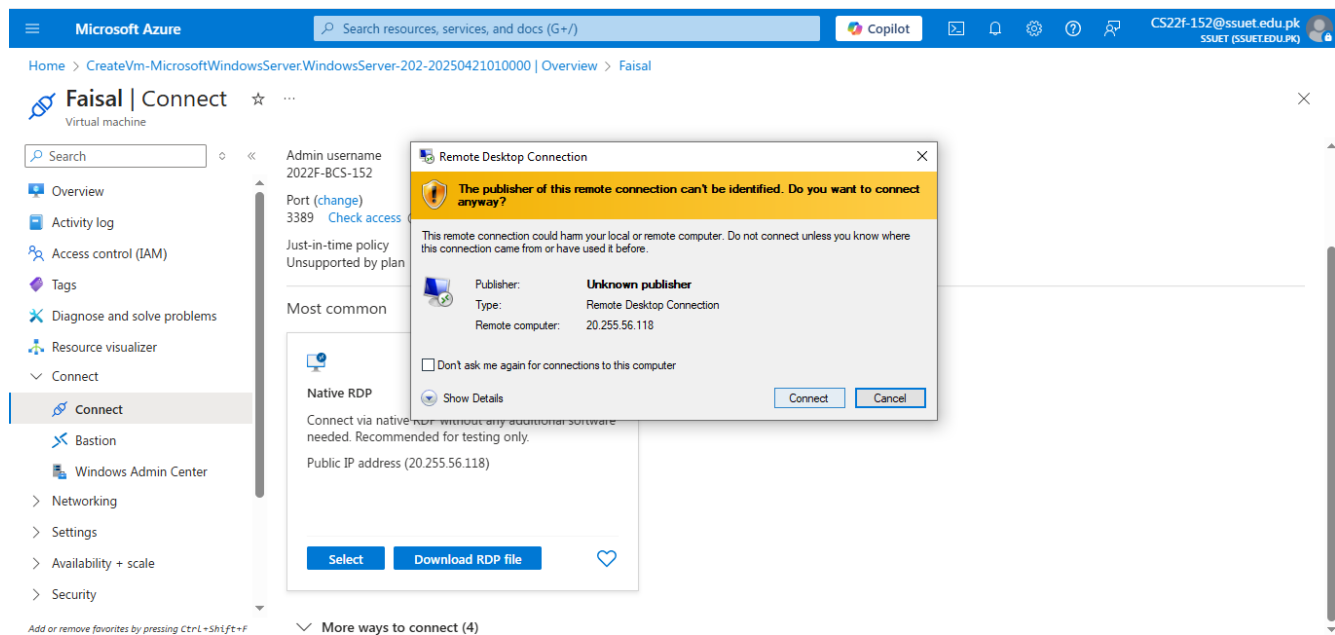
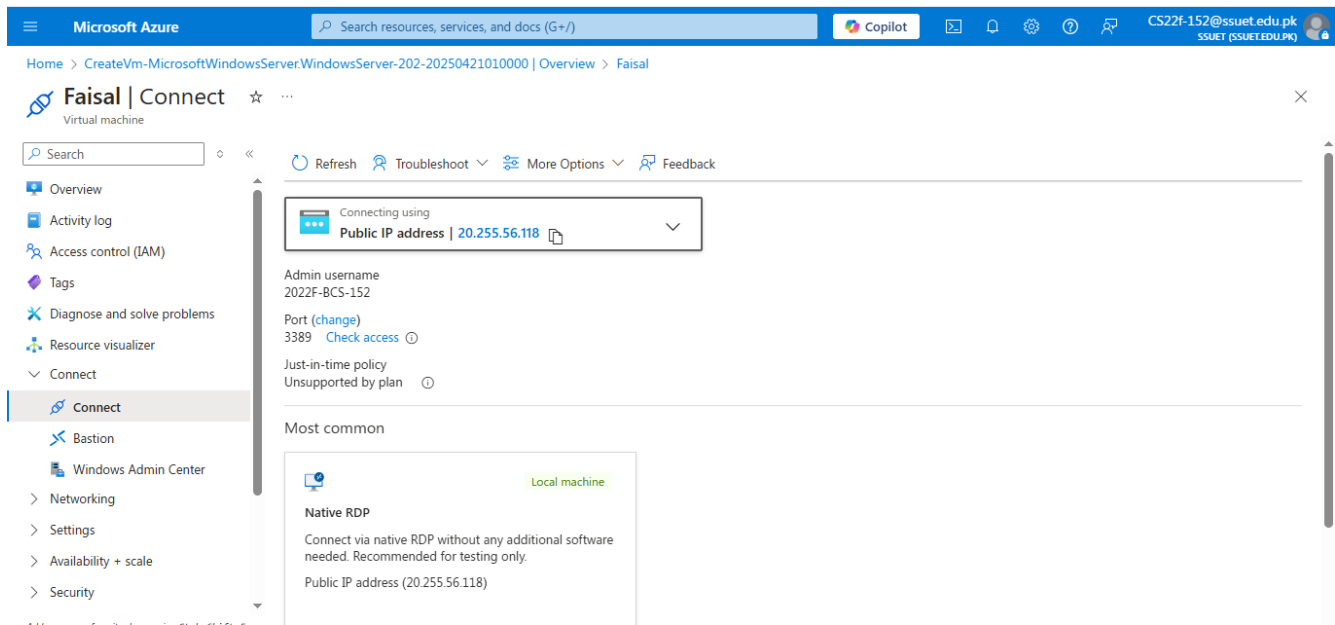
View account

Switch directory

Sign in with a different account

JSON View

✓ Host machine connecting to the cloud VM



Faisal | Connect

Virtual machine

Search

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Connect

Connect

Bastion

Windows Admin Center

Networking

Settings

Availability + scale

Security

Admin username
2022F-BCS-152

Port (change)
3389 [Check access](#)

Just-in-time policy
Unsupported by plan

Most common

Local machine

Native RDP

Connect via native RDP without any additional software needed. Recommended for testing only.

Public IP address (20.255.56.118)

Select

Download RDP file

More ways to connect (4)

Remote Desktop Connection

Connecting to:
20.255.56.118:3389

Securing remote connection...

Cancel